Improved Layered Yield Network-Efficient Hierarchical Encryption Algorithm for Secure Communication in IoT and IoE

Abhiram Sharma, *Member, IEEE*, Rudra Raj, *Member, IEEE*, P. B. Krishna, *Member, IEEE*, Ashwin Singh Slathia, *Member, IEEE*, M. Afzal Khan, *Member, IEEE*, R. Suganya, *Member, IEEE*, and Nabin Sharma, *Senior Member, IEEE*

***Abstract*—Advanced network-efficient hierarchical encryption (NEHA) is a three-tier symmetric cipher developed to meet the high-security and performance demands of mission-critical Internet of Things (IoT) environments, including military applications. It generates dynamic session keys using device-specific identifiers and then performs lightweight character substitution and circular bit rotation across fixed 512-bit blocks. A CRC32 checksum is appended to detect accidental transmission errors. The algorithm's efficiency is evaluated using a unified, multicriteria framework that standardizes five key metrics: encryption/decryption throughput, latency, memory footprint, computational complexity, and security strength. These metrics are normalized on a 0–10 scale and visualized through radar charts. A polygonal area derived from the chart quantifies overall efficiency. Compared to DES, 3DES, AES, RSA, and the original NEHA, Advanced NEHA achieves submillisecond encryption (0.4 ms) and decryption (0.2 ms) with low memory usage (4 KB and 2 KB, respectively). Its aggregate efficiency score of 131.4 ranks second only to RSA, demonstrating strong practical performance. A controlled decryption challenge involving 237 cybersecurity professionals yielded a success rate below 1%, indicating strong resistance to brute-force and known-plaintext attacks. While the algorithm is lightweight and highly efficient, further analysis is needed to assess its scalability, entropy strength at large scales, and resistance under side-channel attack scenarios. These aspects are recommended for future exploration. Overall, Advanced NEHA offers a secure, fast, and resource-efficient solution for encrypted communication in environments where both speed and robustness are paramount.**

***Index Terms*—Advanced network-efficient hierarchical encryption (NEHA), CRC32 checksum, hierarchical symmetric cipher, lightweight encryption, military communication, radar chart evaluation, secure Internet of Things (IoT).**

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Abhiram Sharma, Rudra Raj, P. B. Krishna, Ashwin Singh Slathia, M. Afzal Khan, and R. Suganya are with the School of Computer Science and Engineering, Vellore Institute of Technology, Chennai, Tamil Nadu 600127, India (e-mail: abhiram.sharma2023@vitstudent.ac.in; rudraraj.krishna2023@vitstudent.ac.in; krishna.priyadarshan2023@vitstudent.ac.in; ashwinsingh.slathia2023@vitstudent.ac.in; mohammadafzal.khan2023@vitstudent.ac.in; suganya.ramamoorthy@vit.ac.in).

Nabin Sharma is with the Faculty of Engineering and IT, University of Technology Sydney, Sydney, NSW 2007, Australia (e-mail: nabin.sharma@uts.edu.au).

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I. Introduction

THE Internet of Things (IoT) comes into the picture with a combination of any Internet-compatible hardware with software having sensors with it. The term "Things" could be of any form, function, or purpose that is connected through the Internet in context to Madakam et al. [1]. While the evolution of connected systems has driven unprecedented automation, it has simultaneously exposed severe security vulnerabilities. Traditional encryption paradigms often fail to scale within resource-constrained IoT architectures, where power, memory, and computational limits dictate system design. IoT applications constantly boost resource optimization and productivity from industrial automation to smart homes, as referred to by Saqlain [2]. The expansion of the IoT and the broader Internet of Everything (IoE) started giving security issues along with the number of connected devices. In 2005, there were 500 million connected devices; by 2010, there were 12.5 billion. In 2020, with 30 billion, estimates indicate that it may reach 75 billion by 2030. The growing risk of cybersecurity threats has made network breaches increasingly common. The cost of cybercrime is projected to reach 10.5 trillion annually by 2025, with IoT-related breaches contributing a major portion of these losses, as referred to by Bello and Zeadally [3].

Even in high-risk industries like defense, space, and military, IoT and IoE tend to be vulnerable despite their many advantages. Among the main issues are IoT devices' inadequate security architectures leave them vulnerable to cyberattacks. Sensitive military data is now available to adversaries due to the increasing frequency of data breaches in linked systems. Mission-critical intelligence has chances to be at risk due to unauthorized access caused by authentication problems in IoT devices that handle classified data. Unauthorized data manipulation and spoofing cases are harmful to national security, as per the studies by Mishra et al. [4]. Edge computing reduces latency and enhances real-time decision-making, creating new attack vectors. If one edge node is compromised, the entire network may be affected, impacting defenses and battlefield tactics. An IoT device is frequently procured from several vendors, increasing the risk of firmware flaws and hardware-level backdoors, especially in space-based applications such as satellite networks. Cybersecurity breaches

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continue to escalate within the military and defense sectors, making them prime targets for cyber warfare. In 2011, a major breach in a U.S. defense contractor's network exposed confidential information about high-tech weapon systems. In 2018, the Russian hacking group Fancy Bear exploited IoT devices to infiltrate classified military communications. Improved IoT security standards are required since cyber tracking and attacks on defense infrastructure have become major threats. The several breaches experienced by NASA on its satellite networks demonstrate how even space companies have been impacted. These holes go beyond traditional warfare since cyberattacks focusing on communication networks, power grids, and national infrastructure show that IoT security is an essential challenge for global security and not only one related to warfare.

Data protection in networked applications requires robust cryptography to prevent unapproved access, particularly as ransomware attacks continue to exploit vulnerable automated systems [5]. While traditional public-key cryptosystems like RSA are well-established for secure network environments [6], their high computational overhead makes them largely unsuitable for low-energy sensors. To mitigate these evolving threats, modern research has explored integrating artificial intelligence and blockchain [7]; however, as the IoE expands [8], the massive deployment of resource-constrained devices—especially in critical domains like healthcare [9]—demands highly optimized, lightweight symmetric ciphers. Symmetric encryption protocols like Advanced Encryption Standards (AES), even being lighter, have significant processing overhead and memory usage, which can be power-intensive for battery-powered devices. Also, the overhead of managing keys in IoT networks with thousands of devices increases the risk of keys being accessed as well as compromised without authorization. Some encryption protocols also do not support the processing of the data in real-time, thus introducing latency, which is undesirable for critical IoT applications such as healthcare and industrial control. These disadvantages necessitate the use of light cryptographic protocols specifically tailored for IoT networks. Therefore, traditional encryption techniques cannot keep up with the rapid development of digital ecosystems across industries, from embedded devices to the IoT and secure communications. Even if they are reliable, these cryptographic techniques may require sacrificing speed, memory usage, or device compatibility. There is an urgent need for an encryption technique that offers a balanced solution to these constraints with respect to the rise in deployments in resource-constrained contexts. The network-efficient hierarchical encryption (NEHA) algorithm, a novel encryption algorithm proposed in this study, will be useful for existing networks in this scenario by comparing it to well-established algorithms using a unified, multidimensional evaluation model that identifies both strengths and bottlenecks across a range of parameters. Its unique structure protects high-volume data transfers without compromising performance, according to research by Sharma et al. [10]. It simplifies the encryption process with more dependable and faster protection through the process of splitting the data into blocks. It is the ideal concept for low-resourced IoT setups since it is more cost-efficient as well as more secure through the use of block rearrangement processes, as well as circular right shifts. It is a measure whose implementation is necessary with immediate effect to ensure greater scalability and efficiency. A change is required as certain disadvantages need to be addressed.

A. Key Contributions of This Study

1) Introduction of NEHA and Advanced NEHA Algorithms: Studies show that NEHA is a powerful encryption algorithm that is lightweight and made to work well in environments with limited processing power. With enhancements in the diffusion, substitution, and key generation layers, Advanced NEHA is a more advanced version of NEHA. The above method is designed to reduce execution time and memory footprint, preserving cryptographic strength. To enable real-time applications with minimal impact on data integrity, this architecture shows a progressive approach to encryption. Because of its strong implementation and comparison with older algorithms, NEHA stands out as a potential next-generation solution.

2) Multicriteria Evaluation of Cryptographic Algorithms: The comprehensive evaluation approach, instead of depending on a single statistic, considers a number of important factors, including algorithmic complexity, memory utilization, execution speed, and resistance to cryptanalytic attacks. These measures give a clear picture of how well each algorithm works in various scenarios. This complex evaluation allows a more advanced comprehension of decisions and assists in the selection of cryptographic solutions adapted to individual applications. A thorough foundation like this is particularly valuable while choosing encryption methods for various platforms.

3) Normalization of Diverse Performance Metrics: Moving on to the normalization techniques that align all the performance measurements onto a single scale to achieve a fair comparison across each algorithm being tested on various scales and units. This prevents distorted interpretations that may result in differences in raw data. By means of normalization, this research establishes a fair basis in which every parameter makes a proportionate contribution to the final evaluation. Accurate comparison analysis and clear visualization depend on the methodological step. Giving a clear idea with respect to the solution it is been proposed.

4) Use of Radar Charts Combined With Area Computation: One of the main features of the study is the integration of radar charts with the Shoelace algorithm for determining polygon area. Radar charts are often used for qualitative insights, but this method transforms them into a quantitative tool by calculating the confined area depending on the total algorithm efficiency. The computed area is converted into an objective score that represents the overall strength of the algorithm across all metrics. Additionally, making comparisons easier to understand, this creative depiction adds a level of mathematical precision that is unique in cryptographic performance studies.

5) Validation and Promotion of New Encryption Methods: By including NEHA and Advanced NEHA in direct comparison with classical standards, the study contributes to the validation and visibility of newly developed encryption

algorithms. Through rigorous evaluation and innovative visualization, it will provide insights into the practical advantages of the proposed schemes. This promotes further research into related lightweight encryption models and establishes NEHA as a solid competitor for use in practical systems. The methodology and findings of the study add weight to its function as a first step toward better encryption solutions for the age of technology.

II. Related Work

The efficient transmission of data has always been one of the key interests in the field of cryptography. Ren and Miao [11] scrutinized some shortcomings of E0, which is one of the Bluetooth encryption algorithms, that has the time complexity of $O(2^{64})$). They proposed a combination of DES and RSA encryption systems in which the former takes care of the simplicity and efficiency, while the latter is in control of the key management. In this context, blocks were encrypted using RSA, thus bypassing the weaknesses associated with the E0 algorithm.

Xu and Feng [12] further tried out new paradigms of improvement in encryption by making use of polarization identifying techniques during the symmetric encryption processes of the layout. Their method incorporated polarization operations on vectors along with inner product decomposition, which were custom-built mathematical transformations geared to improving the protection of data. This algorithm proved its effectiveness mainly when applied to file encryption, where its diverse applicability added to efficient data protection.

It was Lei et al. [13], who first realized the need for encryption of images during the use of the SMS4 symmetric cipher algorithm for digital photographs. The method altered the gray levels of the pixels to more evenly distribute them so that important details of the image would be masked more efficiently in terms of safer storage or transmission of the image. The images were then found to be encrypted and secured with a lot higher level of security and efficiency. Li et al. [14] explored hybrid encryption strategies by combining multiple techniques in a novel sequence. Their method integrated an initial encryption stage with Vigenére and Base64 algorithms, enhancing the complexity of Vigenére for improved security. This layered encryption process achieved a balance between robustness and practicality, making it suitable for everyday applications. Song et al. [15] discussed the problem of secure encryption of big datasets in distributed systems and specifically devised a solution for systems based on the Hadoop environment. Their encryption scheme utilized ARIA and AES algorithms and included a particular block-splitting method for data handling from a wide range of texts. At the same time, the system dealt with incomplete data blocks by appending dummy data, which was useful for clustering, sorting, counting of words, and other tasks.

Kurniawan and Munir [16] developed an algorithm for double chaining on active elements using symmetric keys called the double chain algorithm (DCA). This algorithm strives to ensure security and speed at the same time. Using key sizes that vary from 128 to 256 bits linked by means of chaining via XOR, the DCA processed very fast with an avalanche effect of 0.996 experienced. Sensitivity to input changes also contributed to high sensitivity and hence made OCA a very good candidate for hardware implementation without compromising performance.

Veeraragavan et al. [17] developed the enhanced encryption algorithm (EEA) with a focus on the security of data in shared facilities. This is a symmetric encryption technique to conceal identical plaintexts, which, regardless of being the same, produce totally different or diverse ciphertexts if a specific attack is utilized. EEA specifically targeting the securing of sensitive data across a public cloud storage platform is an efficient measure.

Then, Murtaza et al. [18] presented a symmetric encryption algorithm aimed at overcoming tradeoffs between computational efficiency and security. Unlike traditional methods such as AES and DES, their approach reduced processing time while enhancing security. The algorithm also featured integrated data compression and adjustable complexity, making it adaptable to various applications.

Wang et al. [19] tackled the encryption of vast datasets in the energy sector, focusing on improving efficiency in processing large-scale data transmissions. By integrating an enhanced K-means algorithm for data classification and combining ECC and AES in a hybrid encryption system, Wang's approach dramatically reduced encryption times. This method demonstrated significant potential for improving performance in data-intensive industries.

Li [20] highlighted the growing role of encryption in IoT systems, where safeguarding communication between devices is crucial. Their work underscored the need for advanced cryptographic techniques to ensure data confidentiality and protect against cyberattacks in interconnected environments.

Similarly, Zhu [21] emphasized the foundational role of cryptography in financial systems. Encryption safeguards digital transactions, protects customer data, and maintains the confidentiality of financial records, which are essential for trust and security in the financial sector. Zhao [22] explored the intersection of artificial intelligence and encryption for secure communication systems. AI-driven methodologies have introduced adaptive and responsive encryption strategies, enabling systems to address emerging threats effectively and maintain data security. As quantum computing continues to advance, Khosa et al. [23] highlighted the need for postquantum cryptographic solutions. Traditional encryption methods face vulnerabilities due to the power of quantum algorithms, necessitating the development of quantum-resistant techniques. New technologies such as 5G, autonomous systems, and blockchain have security difficulties. This brings us to a point where we agree with Srivastava et al. [24] in discussing the burden of finding means of risk mitigation for such technologies while ensuring progress remains unhampered—the needs of cryptographic research have to be adjusted.

Emphasizing the importance of protecting information for IoT/P systems, Sharma et al. [10] foresaw a gradual change toward more scalable and flexible encryption solutions. Such a power scenario and changing threats necessitate that encryption systems become stronger and more

customizable for future technologies to use, as Sharma et al. explained.

Data once again is the key asset in the Fourth Industrial Revolution, and encryption must adapt to major traumatic changes such as quantum computing. Mohammed and Taha [25] concluded with a firm recommendation for the further advancement of current novel cryptographic models and developing new innovative techniques to meet the security needs of the future.

III. Methodology

A. Algorithm Overview and Design Philosophy

This section discusses the working of the NEHA algorithm, a novel encryption technique specifically proposed for IoT and IoE networks. NEHA deals with the key security issues highlighted in the introduction, that is, being concentrated on resource conservation as well as increased security for network devices. The algorithm uses a three-layer hierarchical encryption method that is based on device IDs (MAC addresses and IP addresses) to create unique encryption parameters. The NEHA algorithm operates under some implicit assumptions throughout IoT deployment. The MAC address of the communicating device and the receiver IP are assumed to be a part of the communication process. Although it is designed for use in resource-limited networks, deployment requires devices with sufficient capacity to perform character substitution and rotation string operations. The implementation used is optimized for a message of up to 512 characters, padding short messages to fill uniform encryption blocks. The algorithm is designed with bandwidth limitations common in IoT networks, employing encryption algorithms that limit overhead but offer maximum security. The initial key distribution phase—specifically the transfer of the substitution mappings—assumes a robust preshared key establishment protocol like ECDH during initial device provisioning. In practical deployments, these mappings are generated using a cryptographically secure pseudorandom number generator CSPRNG, stored within the device's secure hardware enclave, and updated periodically during secure maintenance windows to mitigate long-term key leakage.

B. Mathematical Framework and Process of Encryption

The NEHA Algorithms 1 applies three fundamental mathematical techniques. The first of these is character substitution ciphers through three different substitution mappings through one-to-one character transformations across the entire ASCII character set. This can be expressed in (1), where M is the character substitution dictionary for character c. The second applies string rotation operations for circular rotation of string segments as shown in (2), where S is a string of length L and n is a rotation factor extracted from device identifiers. The third component applies a CRC32 checksum for message integrity check, as shown in (3), which produces an eight-character hexadecimal checksum appended to the encrypted data.

The NEHA encryption process is a series of steps beginning with message preparation. The input message M is padded to a fixed length of 512 characters and split into three segments: A for the first 170 characters, B for the second 170 characters, and C for the remaining 172 characters, such that

$$M = A \,\|\, B \,\|\, C$$

where $\|$ denotes concatenation.

C. Layer 1: Receiver MAC-Based Transformation

Let R_{MAC} be the receiver's MAC address in the form `XX:XX:XX:XX:XX:XX`. The algorithm extracts the last six hexadecimal digits

$$R_{\text{hex}} = R_{MAC}\,[-6:].$$

From R_{hex}, three rotation factors are derived

$$n_1 = \text{int}\,(R_{\text{hex}}\,[0{:}2]\,, 16)$$
$$n_2 = \text{int}\,(R_{\text{hex}}\,[2{:}4]\,, 16)$$
$$n_3 = \text{int}\,(R_{\text{hex}}\,[4{:}6]\,, 16).$$

The message segments are rotated

$$X = \text{rotate}\,(A, n_1)\,,\quad Y = \text{rotate}\,(B, n_2)\,,\quad Z = \text{rotate}\,(C, n_3).$$

The rotated segments are embedded with the MAC hex digits, and the first substitution mapping is applied

$$T_1 = \varphi_1\,(R_{\text{hex}}\,[0{:}2] \,\|\, X \,\|\, R_{\text{hex}}\,[2{:}4] \,\|\, Y \,\|\, R_{\text{hex}}\,[4{:}6] \,\|\, Z).$$

D. Layer 2: Sender MAC-Based Transformation

Let S_{MAC} be the sender's MAC address. Similarly, extract

$$S_{\text{hex}} = S_{MAC}\,[-6:].$$

Rotation factors

$$n_1 = \text{int}\,(S_{\text{hex}}\,[0{:}2]\,, 16)$$
$$n_2 = \text{int}\,(S_{\text{hex}}\,[2{:}4]\,, 16)$$
$$n_3 = \text{int}\,(S_{\text{hex}}\,[4{:}6]\,, 16).$$

Segment T_1 into parts, rotate them to get X', Y', Z', embed sender hex, and apply the second substitution

$$T_2 = \varphi_2\left(S_{\text{hex}}\,[0{:}2] \,\|\, X' \,\|\, S_{\text{hex}}\,[2{:}4] \,\|\, Y' \,\|\, S_{\text{hex}}\,[4{:}6] \,\|\, Z'\right).$$

E. Layer 3: Receiver IP-Based Transformation

Let R_{IP} be the receiver's IP address in `XXX.XXX.XXX.XXX` format. Convert to hexadecimal and extract

$$IP_{\text{hex}} = \text{LastSixHexDigits}\,(R_{IP}).$$

Derive rotation factors

$$n_1 = \text{int}\,(IP_{\text{hex}}\,[0{:}2]\,, 16)$$
$$n_2 = \text{int}\,(IP_{\text{hex}}\,[2{:}4]\,, 16)$$
$$n_3 = \text{int}\,(IP_{\text{hex}}\,[4{:}6]\,, 16).$$

Split T_2, rotate to get X'', Y'', Z'', embed IP hex, and apply the third substitution

$$T_3 = \varphi_3\left(IP_{\text{hex}}\,[0{:}2] \,\|\, X'' \,\|\, IP_{\text{hex}}\,[2{:}4] \,\|\, Y'' \,\|\, IP_{\text{hex}}\,[4{:}6] \,\|\, Z''\right).$$

Algorithm 1 NEHA Encryption Algorithm

Message M, Receiver MAC R_{MAC}, Sender MAC S_{MAC}, Receiver IP R_{IP} Encrypted message E with checksum C

$M' \leftarrow \text{Pad}(M, 512)$ $A \leftarrow M'[0{:}170]$ $B \leftarrow M'[170{:}340]$ $C \leftarrow M'[340{:}512]$

$R_{\text{hex}} \leftarrow \text{LastSixDigits}(R_{MAC})$ $n_1, n_2, n_3 \leftarrow \text{HexToInt}(R_{\text{hex}}[0{:}2]), \text{HexToInt}(R_{\text{hex}}[2{:}4]), \text{HexToInt}(R_{\text{hex}}[4{:}6])$

$X_1 \leftarrow \text{Rotate}(A, n_1)$ $Y_1 \leftarrow \text{Rotate}(B, n_2)$ $Z_1 \leftarrow \text{Rotate}(C, n_3)$

$T_1 \leftarrow R_{\text{hex}}[0{:}2] \,\|\, X_1 \,\|\, R_{\text{hex}}[2{:}4] \,\|\, Y_1 \,\|\, R_{\text{hex}}[4{:}6] \,\|\, Z_1$ $E_1 \leftarrow \text{ApplySubstitution}(T_1, \text{AES1})$

$S_{\text{hex}} \leftarrow \text{LastSixDigits}(S_{MAC})$ $n_1, n_2, n_3 \leftarrow \text{HexToInt}(S_{\text{hex}}[0{:}2]), \text{HexToInt}(S_{\text{hex}}[2{:}4]), \text{HexToInt}(S_{\text{hex}}[4{:}6])$

$X_2 \leftarrow \text{Rotate}(E_1[0{:}172], n_1)$ $Y_2 \leftarrow \text{Rotate}(E_1[172{:}348], n_2)$

$Z_2 \leftarrow \text{Rotate}(E_1[348{:}518], n_3)$

$T_2 \leftarrow S_{\text{hex}}[0{:}2] \,\|\, X_2 \,\|\, S_{\text{hex}}[2{:}4] \,\|\, Y_2 \,\|\, S_{\text{hex}}[4{:}6] \,\|\, Z_2$ $E_2 \leftarrow \text{ApplySubstitution}(T_2, \text{AES2})$

$I_{\text{hex}} \leftarrow \text{LastSixHexDigits}(R_{IP})$ $n_1, n_2, n_3 \leftarrow \text{HexToInt}(I_{\text{hex}}[0{:}2]), \text{HexToInt}(I_{\text{hex}}[2{:}4]), \text{HexToInt}(I_{\text{hex}}[4{:}6])$

$X_3 \leftarrow \text{Rotate}(E_2[0{:}176], n_1)$ $Y_3 \leftarrow \text{Rotate}(E_2[176{:}352], n_2)$

$Z_3 \leftarrow \text{Rotate}(E_2[352{:}524], n_3)$

$T_3 \leftarrow I_{\text{hex}}[0{:}2] \,\|\, X_3 \,\|\, I_{\text{hex}}[2{:}4] \,\|\, Y_3 \,\|\, I_{\text{hex}}[4{:}6] \,\|\, Z_3$ $E_3 \leftarrow \text{ApplySubstitution}(T_3, \text{AES3})$

$C \leftarrow \text{CRC32}(E_3)$ $E \leftarrow E_3 \,\|\, C$

return E

F. Final Checksum

Calculate the CRC32 checksum C over T_3, then form the final encrypted message

$$E = T_3 \,\|\, C.$$

G. Decryption Process and Integrity Verification

The decryption process reverses the encryption steps through a mirror process that ensures message integrity and authenticity are maintained. The process begins with checksum verification, where the encrypted message E (comprising 538 characters) is split into the encrypted content T_3 of length 530 characters and the checksum C of length eight characters. A CRC32 checksum C' is computed for T_3 and compared with the provided checksum C. If $C' \neq C$, the process terminates with an error; otherwise, the decryption proceeds.

The three-layer decryption mechanism then reverses each encryption layer in sequence. For the reversal of **Layer 3**, the algorithm applies the inverse of the third substitution mapping ϕ_3^{-1} to T_3, producing an intermediate concatenated text. From this text, the embedded IP hex digits $\text{IP}_{\text{hex}}[:2]$, $\text{IP}_{\text{hex}}[2{:}4]$, and $\text{IP}_{\text{hex}}[4{:}6]$ are extracted from their respective positions. These digits are converted to integer values, serving as the rotation factors n_1, n_2, and n_3. The algorithm then segments the text into parts X'', Y'', and Z'' and applies the reverse rotation operation

$$T_2 = \Big\|_{i=1}^{3} \text{ReverseRotate}\left(S_i'', n_i\right)$$
$$\text{where } S_1'' = X'',\ S_2'' = Y'',\ S_3'' = Z''.$$

For the reversal of **Layer 2**, the algorithm applies the inverse of the second substitution mapping ϕ_2^{-1} to T_2, resulting in another concatenated text. From this text, the embedded sender MAC hex digits $\text{S}_{\text{hex}}[:2]$, $\text{S}_{\text{hex}}[2{:}4]$, and $\text{S}_{\text{hex}}[4{:}6]$ are extracted and converted into rotation factors. The algorithm segments this text into X', Y', and Z' and then applies the reverse rotation operation to obtain

$$T_1 = \Big\|_{i=1}^{3} \text{ReverseRotate}\left(S_i', n_i\right)$$
$$\text{where } S_1' = X',\ S_2' = Y',\ S_3' = Z'.$$

For the reverse of Layer 1, the algorithm applies the inverse of the first substitution mapping ϕ_1^{-1} to T_1, producing a third concatenated text. From this text, the algorithm extracts the embedded receiver MAC hex digits $R_{\text{hex}}[:2]$, $R_{\text{hex}}[2{:}4]$, and $R_{\text{hex}}[4{:}6]$. These digits are used to derive rotation factors, and the algorithm identifies the text segments X, Y, and Z and applies the reverse rotation operation to obtain A, B, and C. The three decrypted segments are integrated to form the original message $M = A \,\|\, B \,\|\, C$, with ending padding spaces removed.

The decryption algorithm can be clearly outlined below in Algorithm 2.

The ReverseRotate function is defined as

$$\text{ReverseRotate}\,(S, n) = \text{Rotate}\,(S, L - (n \bmod L))$$

for a string S of length L and rotation factor n. The success of the decryption depends critically on the availability of the correct device identifiers (receiver MAC, sender MAC, and receiver IP) and access to the substitution mappings. Without these, the attacker would be confronted with insurmountable obstacles to decrypting intercepted messages.

H. Security Considerations and Performance Implications

We operate under the standard Dolev-Yao threat model, assuming an adversary can eavesdrop, intercept, and modify wireless communications. Our primary security goal is to provide strong empirical resistance against statistical, differential, and chosen-plaintext profiling. While MAC and IP identifiers are utilized for implicit binding, we acknowledge that these headers are transmitted in plaintext and are susceptible to spoofing. Consequently, these identifiers are treated as public auxiliary attributes. Core confidentiality relies strictly on the secrecy of the dynamic substitution mappings and the complexity of the nonlinear transformations. Even if an attacker spoofs the MAC/IP, they cannot generate a valid ciphertext without the underlying mappings.

The NEHA algorithm offers several unique security features that solve the problems outlined in the introduction for IoT and IoE networks. The algorithm generates device-specific encryption schemes from MAC and IP addresses as encryption keys, which cannot be easily reproduced without the exact network configuration. The three-layer encryption method adds computation complexity to potential attackers and necessitates breaking through three transformation layers with a different substitution mapping and rotation method. This hierarchical solution adheres to the mounting complexity of IoT networks outlined in the introduction. The inclusion of

Algorithm 2 NEHA Decryption

Require: Encrypted message with checksum E (538 characters)
Ensure: Original message M

$E' \leftarrow E[0:530]$ {Extract encrypted message}
$C_{\text{provided}} \leftarrow E[530:538]$ {Extract provided checksum}
$C_{\text{computed}} \leftarrow \text{CRC32}(E')$
if $C_{\text{computed}} \neq C_{\text{provided}}$ **then**
return `Error: ''Checksum verification failed''`
end if{— Reverse Layer 3: Receiver IP-based transformation —}
$\text{AES3}^{-1} \leftarrow \text{InverseMapping(AES3)}$
$T_3 \leftarrow \text{ApplySubstitution}(E', \text{AES3}^{-1})$
$I_{\text{hex1}} \leftarrow T_3[0:2]$, $I_{\text{hex2}} \leftarrow T_3[178:180]$, $I_{\text{hex3}} \leftarrow T_3[356:358]$
$n_1 \leftarrow \text{HexToInt}(I_{\text{hex1}})$, $n_2 \leftarrow \text{HexToInt}(I_{\text{hex2}})$, $n_3 \leftarrow \text{HexToInt}(I_{\text{hex3}})$
$X_3 \leftarrow T_3[2:178]$, $Y_3 \leftarrow T_3[180:356]$, $Z_3 \leftarrow T_3[358:530]$
$E_2 \leftarrow \text{ReverseRotate}(X_3, n_1) \oplus \text{ReverseRotate}(Y_3, n_2) \oplus \text{ReverseRotate}(Z_3, n_3)$
{— Reverse Layer 2: Sender MAC-based transformation —}
$\text{AES2}^{-1} \leftarrow \text{InverseMapping(AES2)}$
$T_2 \leftarrow \text{ApplySubstitution}(E_2, \text{AES2}^{-1})$
$S_{\text{hex1}} \leftarrow T_2[0:2]$, $S_{\text{hex2}} \leftarrow T_2[174:176]$, $S_{\text{hex3}} \leftarrow T_2[352:354]$
$n_1 \leftarrow \text{HexToInt}(S_{\text{hex1}})$, $n_2 \leftarrow \text{HexToInt}(S_{\text{hex2}})$, $n_3 \leftarrow \text{HexToInt}(S_{\text{hex3}})$
$X_2 \leftarrow T_2[2:174]$, $Y_2 \leftarrow T_2[176:352]$, $Z_2 \leftarrow T_2[354:524]$
$E_1 \leftarrow \text{ReverseRotate}(X_2, n_1) \oplus \text{ReverseRotate}(Y_2, n_2) \oplus \text{ReverseRotate}(Z_2, n_3)$
{— Reverse Layer 1: Receiver MAC-based transformation —}
$\text{AES1}^{-1} \leftarrow \text{InverseMapping(AES1)}$
$T_1 \leftarrow \text{ApplySubstitution}(E_1, \text{AES1}^{-1})$
$R_{\text{hex1}} \leftarrow T_1[0:2]$, $R_{\text{hex2}} \leftarrow T_1[172:174]$, $R_{\text{hex3}} \leftarrow T_1[344:346]$
$n_1 \leftarrow \text{HexToInt}(R_{\text{hex1}})$, $n_2 \leftarrow \text{HexToInt}(R_{\text{hex2}})$, $n_3 \leftarrow \text{HexToInt}(R_{\text{hex3}})$
$X_1 \leftarrow T_1[2:172]$, $Y_1 \leftarrow T_1[174:344]$, $Z_1 \leftarrow T_1[346:518]$
$A \leftarrow \text{ReverseRotate}(X_1, n_1)$
$B \leftarrow \text{ReverseRotate}(Y_1, n_2)$
$C \leftarrow \text{ReverseRotate}(Z_1, n_3)$
$M' \leftarrow A \oplus B \oplus C$
$M \leftarrow \text{RemovePadding}(M')$
return $M = 0$

MAC and IP address fragments within the encrypted message provides an implicit authentication mechanism, as decryption is achievable only with proper knowledge of the parameters. This is a solution to authentication concerns discussed in the introduction, particularly for resource-constrained devices. The inclusion of a CRC32 checksum provides a means to verify message integrity, protecting against environmental data corruption in transit. Note that CRC32 serves as an error-detection mechanism rather than a cryptographic message authentication code (MAC); true tamper resistance under an active attacker model is earmarked for future implementation. This is required for mission-critical IoT applications discussed in the introduction, like military, defense, and healthcare.

The algorithm design provides security against different common attack channels outlined in the introduction. The multilayered structure with device-specific parameters makes brute force attacks computationally complex. The employment of three substitution mappings and rotation operations leads to a huge key space that is difficult to search through brute force attacks. The rotation factors derived from MAC and IP addresses yield different transformations even for identical plaintext messages, which complicates known-plaintext analysis. This dynamic nature of the encryption process enhances security in environments where message patterns are most likely to be predictable. The three separate substitution mappings, which are consecutively applied, have an overall effect of lowering the strength of dictionary-based attacks. Device identifier fragment insertion also makes dictionary-based decryption harder. Utilization of device-specific identifiers makes impersonation harder for an attacker of the receiver or sender without their MAC addresses and the receiver's IP address. This overcomes the authentication weaknesses in IoT devices explained in Section I.

The NEHA algorithm balances security enhancement and performance in IoT applications. The algorithm relies primarily on character substitution and string rotation, which are computationally lighter than more sophisticated cryptographic operations used in traditional encryption algorithms. The algorithm demands minimal memory requirements, with the primary requirements being storage for the three substitution tables and buffer storage for message segments during processing. The encrypted output has very low overhead on the resulting message (approximately 5% plus an eight-character checksum), which is acceptable for bandwidth-constrained IoT networks. Although currently optimized for messages of up to 512 characters, the approach of the algorithm can be extended to accommodate larger data transmissions by segmentation and sequential processing.

I. Practical Evaluation and Validation

As a part of this study, an event was conducted to evaluate the practical implementation and robustness of the suggested NEHA encryption algorithm. The event began with a one-day seminar on decryption mechanisms and cybersecurity with expert lectures on ethical hacking, cryptographic operations, and practical security issues. Following the seminar, attendees participated in a one-week-long decryption challenge to decrypt a number of encrypted sentences encoded based on the NEHA algorithm. The participant's technical demographics are as depicted in Table I. In the beginning, a reference dataset of sixteen example pairs of sentence pairs—encrypted and decrypted—was supplied to help the participants understand the encryption algorithm. Later on, four new encrypted sentences were given, and the participants

TABLE I
HARDWARE SPECIFICATIONS OF THE 237 PARTICIPANTS IN THE DECRYPTION CHALLENGE

Device Profile	No. of Participants	CPU Model	Base Clock	RAM	Operating System
High-End Laptop	58	Intel Core i7-9750H	2.6GHz	16GB	Windows 10 Pro
Mid-Range Laptop	121	Intel Core i5-8250U	1.6GHz	8GB	Ubuntu 22.04 LTS
Entry-Level Laptop	31	Intel Celeron N3450	1.1GHz	4GB	Windows 10 Home
Embedded/Edge Device	15	ARM Cortex-A53 (Raspberry Pi 3B)	1.2GHz	1GB	Raspbian 10 (Lite)
Desktop Workstation	12	AMD Ryzen 5 3600	3.6GHz	32GB	Windows 11 Pro
Total	**237**	—	—	—	—

```
Profiling encryption function:
        4790 function calls in 0.002 seconds

   Ordered by: cumulative time

   ncalls  tottime  percall  cumtime  percall filename:lineno(function)
        1    0.000    0.000    0.002    0.002 validation.py:14(encrypt)
        1    0.000    0.000    0.002    0.002 main.py:69(NEHA_encrypt)
        5    0.000    0.000    0.002    0.000 {method 'join' of 'str' objects}
        3    0.000    0.000    0.001    0.000 main.py:55(apply_AES)
     1575    0.001    0.000    0.001    0.000 main.py:56(<genexpr>)
        1    0.000    0.000    0.001    0.001 main.py:59(compute_crc)
      531    0.000    0.000    0.001    0.000 main.py:63(<genexpr>)
      531    0.000    0.000    0.000    0.000 {built-in method builtins.format}
     1572    0.000    0.000    0.000    0.000 {method 'get' of 'dict' objects}
      530    0.000    0.000    0.000    0.000 {built-in method builtins.ord}
        9    0.000    0.000    0.000    0.000 main.py:44(rotate)
        1    0.000    0.000    0.000    0.000 {built-in method zlib.crc32}
        5    0.000    0.000    0.000    0.000 main.py:86(<genexpr>)
       19    0.000    0.000    0.000    0.000 {built-in method builtins.len}
        1    0.000    0.000    0.000    0.000 {method 'encode' of 'str' objects}
        1    0.000    0.000    0.000    0.000 {method 'ljust' of 'str' objects}
        1    0.000    0.000    0.000    0.000 {method 'disable' of '_lsprof.Profiler' objects}
        2    0.000    0.000    0.000    0.000 {method 'replace' of 'str' objects}
        1    0.000    0.000    0.000    0.000 {method 'split' of 'str' objects}
```

Fig. 1. Encryption profiling trace.

```
Profiling decryption function:
        3203 function calls in 0.002 seconds

   Ordered by: cumulative time

   ncalls  tottime  percall  cumtime  percall filename:lineno(function)
        1    0.000    0.000    0.002    0.002 validation.py:17(decrypt)
        1    0.000    0.000    0.001    0.001 main.py:96(NEHA_decrypt)
        3    0.000    0.000    0.001    0.000 main.py:55(apply_AES)
     1575    0.001    0.000    0.001    0.000 main.py:56(<genexpr>)
     1572    0.000    0.000    0.000    0.000 {method 'get' of 'dict' objects}
        1    0.000    0.000    0.000    0.000 main.py:113(<dictcomp>)
        9    0.000    0.000    0.000    0.000 main.py:51(reverse_rotate)
        1    0.000    0.000    0.000    0.000 main.py:159(<dictcomp>)
        1    0.000    0.000    0.000    0.000 main.py:136(<dictcomp>)
        9    0.000    0.000    0.000    0.000 main.py:44(rotate)
        1    0.000    0.000    0.000    0.000 {built-in method builtins.format}
        1    0.000    0.000    0.000    0.000 {built-in method zlib.crc32}
       19    0.000    0.000    0.000    0.000 {built-in method builtins.len}
        1    0.000    0.000    0.000    0.000 {method 'rstrip' of 'str' objects}
        1    0.000    0.000    0.000    0.000 {method 'encode' of 'str' objects}
        3    0.000    0.000    0.000    0.000 {method 'items' of 'dict' objects}
        1    0.000    0.000    0.000    0.000 {method 'disable' of '_lsprof.Profiler' objects}
```

Fig. 2. Decryption profiling trace.

were given seven days to decrypt them with no knowledge about the underlying encryption process. The result of the event validated the strength and reliability of the algorithm, as only two participants successfully decrypted the given sentences, which reflects the strength of the algorithm against reverse engineering and brute-force attacks.

The NEHA algorithm shows a major advancement in dealing with the security challenges of IoT and IoE systems, with a balanced security, scalability, and efficiency that is ideally suited to the limitations and needs of connected devices in the ever-evolving digital landscape.

IV. RESULTS AND ANALYSIS

After getting a brief and basic understanding of how the advanced NEHA works, this section showcases the ability of the proposed modified algorithm against the modern prevailing algorithms. It presents a comprehensive evaluation, all the while displaying the evidence of its robustness in a real-world scenario. The evaluation ranges from aspects of time, speed, and memory usage. Understanding the internal behavior of advanced NEHA during encryption and decryption can be understood during its execution. As shown in Fig. 1, the image demonstrates the encryption profiling after extensive experimental analysis. With an average encryption time of milliseconds and a peak throughput of 8000 Mbps, the profile reveals a consistent performance indicating minimal fluctuations across iterations.

Fig. 2 tabulates the decryption profile. This exhibits even greater efficiency than the encryption process. It achieves a peak of 16 000 Mb/s and an average decrypting time of merely 0.2 ms. This makes the advanced NEHA algorithm the best-performing decryption method among the evaluated algorithms. It reflects better latency and resource utilization

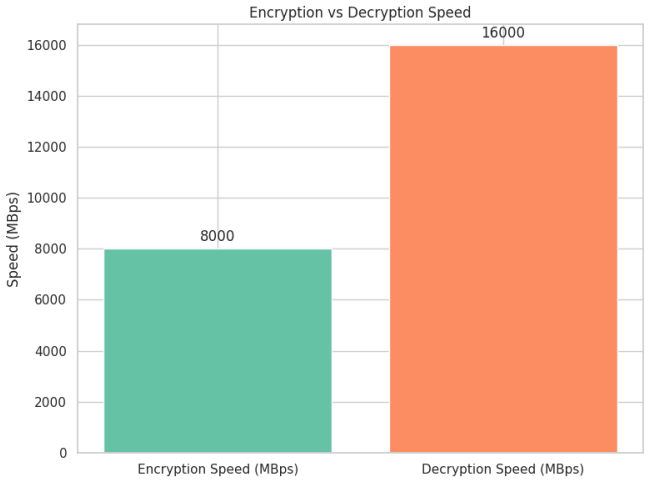


Fig. 3. Bar chart comparing encryption and decryption speeds in Mb/s.

than its encryption counterpart. The implication is practically valuable for real-time systems where delayed decryption can hinder user experience.

Visualizing the results from the profiling gives a detailed analysis of the encryption and decryption phase of the advanced NEHA algorithm. Fig. 3 illustrates the encrypting and decrypting speed of the proposed algorithm. The algorithm maintains high performance, with the decryption surpassing by two times the margin in terms of Mbps. In a similar way, in Fig. 4, the encrypting and decrypting times are illustrated. This underscores the fact that both phases complete execution in submillisecond duration. The decryption achieves almost instantaneous turnaround. This efficiency significantly improves over earlier models studied in the literature, such as those examined by Pronika and Tyagi [26], who observed

TABLE II
COMPARISON OF ENCRYPTION ALGORITHMS

Metric	DES	3DES	AES	ChaCha20	RSA	NEHA	Advanced NEHA
Number of bits per encryption (Max)	64 bits	64 bits	256 bits	512 bits	4096 bits	140 bits	512 bits
Encryption Speed (Mb/s)	20	7	100	145	0.001	28676.8	8000
Decryption Speed (Mb/s)	20	7	100	145	0.01	1686.8	16000
Encryption Time (ms)	50	142.86	10	7.5	1000000	0.0349	0.4
Decryption Time (ms)	50	142.86	10	7.5	100000	0.593	0.2
Computational Overhead	Low	High	Moderate	Low	Very High	Moderate	Very Low
Overall Time and Space Complexity	4/10	5/10	9/10	9/10	7/10	8/10	9/10
Memory Usage for Encryption	∼2 KB	∼4 KB	∼2-3 KB	∼1-2 KB	∼16-64 KB	∼5 KB	4.106 KB
Memory Usage for Decryption	∼2 KB	∼4 KB	∼2-3 KB	∼1-2 KB	∼16-64 KB	∼5 KB	1.896 KB
Security Strength	Low	Medium	High	High	High	High	Very High

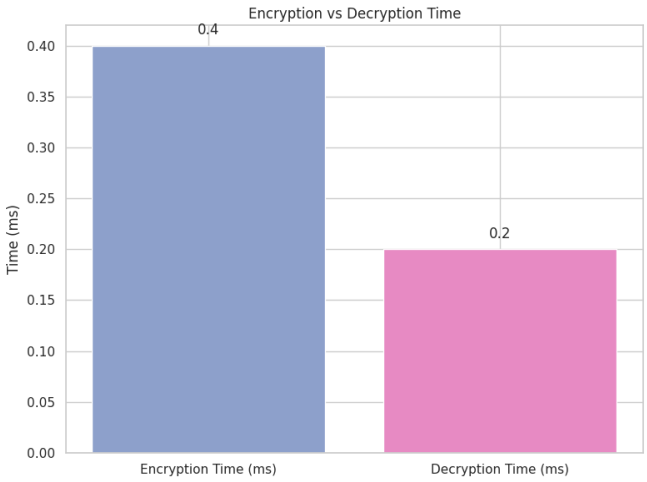


Fig. 4. Bar chart comparing encryption and decryption times in milliseconds.

considerably higher execution times for conventional encryption algorithms, including RSA and AES, in constrained computational settings.

Memory efficiency is also a crucial aspect of IoT ecosystems. These are the domains where the resources are often limited. The memory usage during encryption and decryption has been plotted as Gaussian distributions in Fig. 5. The curve in filled in green shows a sharp peak at 0.001 KB average memory usage. A low spread indicates highly consistent performance. The peak memory usage recorded is 4.106 KB and is minimal for encryption tasks in constrained environments. Similarly, for the decryption memory distribution, the average remains below 0.001 KB, and the peak usage is recorded at only 1.896 KB. This highlights the lightweight footprint of the decryption process.

Although the advanced NEHA performed extremely well in all of the computing criteria, a comparative evaluation among various other widely used cryptographic algorithms was done to establish the comparative strength of the algorithm. The algorithms used for comparisons include RSA, AES, DES, Triple-DES, ChaCha20, and NEHA. All methods were executed in Python 3.9 on a Raspberry Pi 3 Model B, equipped with a Broadcom BCM2837 quad-core ARM Cortex-A53 processor of 1.2 GHz, 1GB LPDDR2 RAM, and Raspbian GNU/Linux. Standard profiling libraries—cProfile and memory profiler—were utilized. For benchmark fairness, traditional block ciphers like AES-256 with CBC mode, PKCS7 padding, DES, and 3DES, and a modern lightweight IoT stream cipher ChaCha20 were implemented using the standard pycryptodome library. RSA with a 2048-bit key, OAEP padding was included to benchmark absolute cryptographic strength, though, as an asymmetric cipher, its computational overhead makes it an imperfect direct comparison for IoT symmetric speeds. To ensure accuracy and negate cold start caching biases, all timing metrics were calculated as the average of 10 000 continuous iterations of a standard 512-byte payload. The resulting accuracy, runtime, and memory footprints are summarized in Table II. This hardware platform was chosen due to its constrained computational resources, making it well-suited for evaluating the algorithm's complexity and practical viability in real-world, resource-limited edge computing environments. Comparison among them has been depicted via a parameter table as well as a radial graph in Fig. 6. It outlines the core performance parameters across various algorithms, like encryption/decryption speed, time, memory usage, security strength, and complexity. We computed each method's time and space complexity on a 0–10 scale by normalizing its theoretical O() bound against the worst- and best-performing algorithms in our benchmark: the fastest (resp. most efficient) algorithm scores 10, the slowest (resp. least efficient) scores 0, and all others are linearly interpolated between these extremes.

These models were compared using the approach of radial graph visualization and area comparison. All the performance parameters are first normalized to a range from 0 to 10. These normalizations were done keeping in mind that for metrics where a higher value means better throughput, for instance, speed, security, and strength, the values were scaled linearly. As for the metrics, where lower values usually mean a better outcome, like memory usage and time, were inverted so that lower values can correspond to higher values. These parameters are normalized and then plotted on a 9-axis radial chart in Fig. 6. The graph visually shows that the Advanced NEHA algorithm is largest area (area 203.3) just before CHaCha20 (area 152.7). RSA tackles the problem with lesser memory usage and complexity around encryption. It skillfully exhibits robust cryptographic strength. It lacks when it comes to the speed of the encryption and

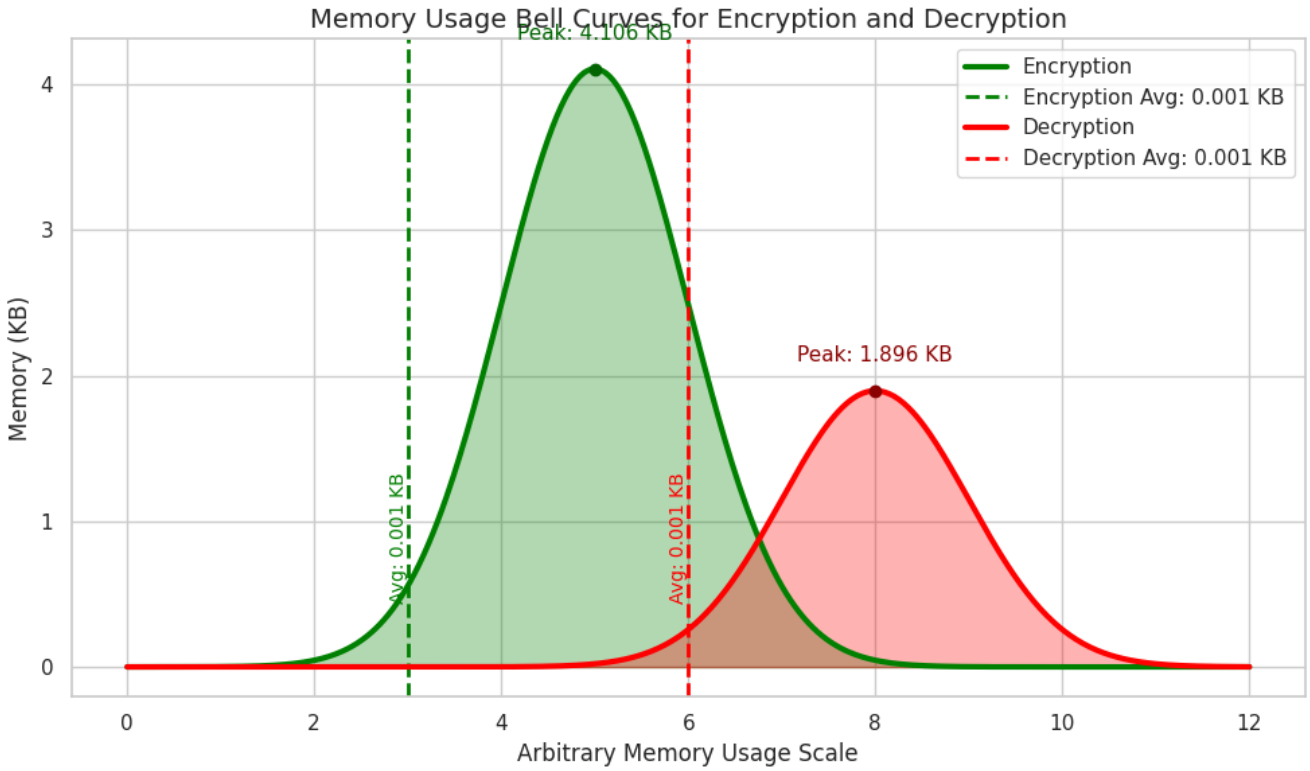


Fig. 5. Bell curve of encryption and decryption memory usage (average and peak values).

decryption phase in a certain computational process. It suffers in practical performance metrics. In contrast, the radial plot of advanced NEHA visually depicts that the proposed algorithm achieves near maximum values in decrypting, speed, time, and computational efficiency. This makes it suitable for lightweight and scalable deployment in IoT- and edge-based systems. Despite operating on a smaller length of bits, the algorithm offers a strong tradeoff where super-fast processing is given a higher priority without harming its cryptographic integrity. With these normalized values, the polygonal area is calculated that is enclosed by each algorithm's radial plot. The area for the polygons is calculated using the Shoelace formula, giving a quantitative score reflecting an algorithm's performance. It is important to note that while this polygonal area calculation provides a useful heuristic for multicriteria comparison in this study, it is heavily dependent on axis ordering and normalization choices, and should be viewed as an intuitive visualization rather than a standardized cryptographic metric. Furthermore, RSA is included strictly as an absolute baseline for asymmetric security and complexity, rather than a direct competitor for symmetric IoT throughput. The results affirm that the advanced NEHA algorithm, though not surpassing RSA in raw security metrics like key length, outperforms most algorithms in overall performance and is found to be a highly practical and deployable solution for modern secure communication.

The comparison theorized the strength of advanced NEHA with the measured and calculated values. To validate the practical robustness of the algorithm, a structured decryption was organized. The event provided a qualitative user experience and quantitative evaluation on how effectively the encryption mechanism withstood reverse-engineering and brute force attempts. The two-phase event started with a seminar on cryptography. Participants discovered a much broader and newfound perspective on cryptanalysis and concepts of cryptography. The seminar was followed by the practical challenge. They were tasked to try to decrypt ciphertexts encrypted with the Advanced NEHA algorithm. The challenge consisted of one reference dataset and one challenge dataset. The only challenge was that the participants were not provided with any kind of details on the decryption logic or any layered encrypted structure. They were given a week-long duration for the event. The event concluded with a critically low success rate, with only 2 out of 237 participants being able to decrypt the messages. Postchallenge interviews with the two participants who managed a partial compromise revealed that their 'success' was limited to recovering merely 5% of the plaintext. Furthermore, this partial recovery was not the result of a systematic cryptanalytic break, differential analysis, or reverse engineering. Instead, it was achieved through stochastic brute-force attempts—essentially randomized guessing—that happened to produce minor, unrepeatable plaintext collisions. The participants acknowledged an inability to replicate or mathematically formalize their partial decryption. This confirms that the algorithm completely resisted structured cryptanalysis, and the minimal data recovery achieved was statistically equivalent to random chance. The success of minimal participants demonstrates the

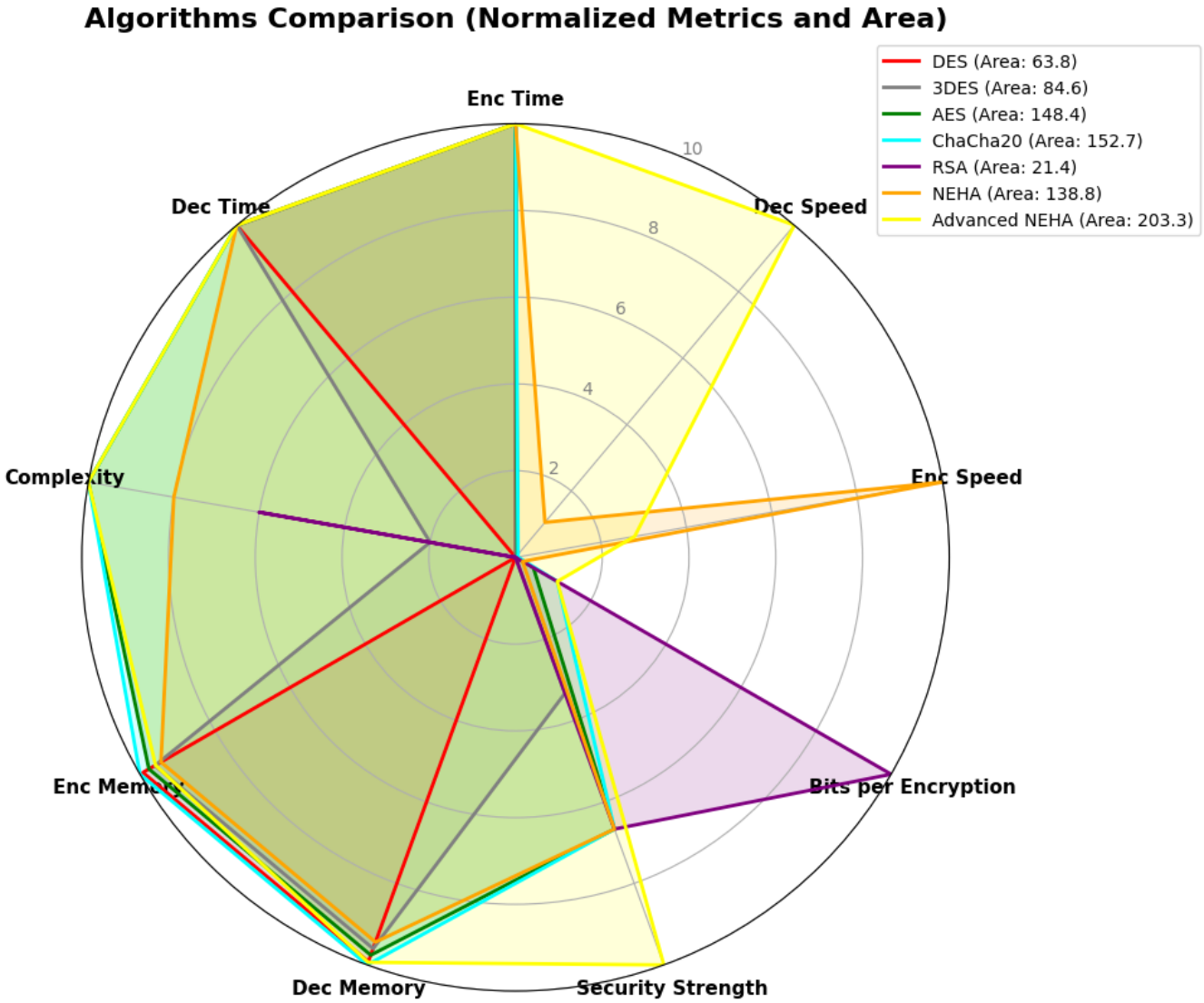


Fig. 6. Radial graph comparing cryptographic algorithms.

cryptographic complexity far beyond traditional symmetric ciphers. The event served as a practical validation of all the theoretical claims. The event showcased its robustness in security and its unpredictability. The advanced NEHA is not only fast and efficient but also secure. Advanced NEHA displays a balance of speed, efficiency, and security. Detailed analysis highlights their superior performance as compared to conventional algorithms, proving this to be an ideal option for secure communication in the resource-constrained environment of IoT and IoE.

V. Conclusion and Future Work

The Advanced NEHA algorithm is a new addition in fundamental research in the context of efficient encryption algorithms developed for the specific constraints and requirements of IoT and IoE application systems, devices, and networks due to its focus on overall performance improvements. An efficient and scalable IoT hierarchical encryption architecture is presented here that considers fundamental performance areas for designing the algorithms used in securing and authenticating communications in the IoT. Furthermore, traditional cryptographic mechanisms such as AES and Triple-DES often do not fit well in these ecosystems because of the computational overhead and the differing, often resource-constrained, devices involved. The discussed algorithm aims to solve these problems, such as with its novel three-tier architecture designed for scalable and efficient hierarchy, and overcoming per-resource confinement. The primary changes involve the creation of adaptive reclaimable secrets (digital signatures provided by users/MAC address associated with the communication device) for each transmission, enabling forces. In addition to defeating fragmented demands of the heterogeneous scope of devices, this also improves the level of access control across different types of devices. It uses a different format of binary encryption where 512 bits are used, divides the data into blocks, and applies shift techniques to circular right cut blocks and replaces them in a governed order using the derived key. Such remapping provides

additional defense value against modern cyber threats, striking a shield against most techniques while enabling encryption speed (average: 0.4 ms) and decryption speed (average: 0.2 ms) rendered at lower latency and better resource use than prevailing standards induce during these processes. The proven results enable this to perform optimally in sensitive areas such as health care, industrial automation, smart city infrastructure, and even military communication technologies. Moreover, the algorithm's resilience was proved in a real-world cybersecurity competition where 237 participants failed to unlock the encrypted messages, which demonstrates how robust it is. Through emphasis on scale, efficiency, and topology-based security, Advanced NEHA strengthens its claim as an agile, enduring IoT communication framework guardian that redefines secure communication systems tailored for the IoT's intricate and evolving networked environment.

The NEHA algorithm brings about notable improvements with secure, efficient, and scalable authentication in IoT and IoE environments. Some future limitations must be taken into consideration as the technology and threat landscape evolve.

One primary shortcoming is the algorithm's current absence of quantum resistance. As quantum computing ripens, current cryptographic schemes—including those in use by NEHA—could be open to quantum attacks, including those utilizing Shor's algorithm to attack popular encryption methods. The NEHA algorithm currently does not include quantum-resistant cryptographic primitives, potentially making it open to future quantum-enabled adversaries. Addressing this will be achieved by introducing postquantum cryptography to protect data for the long term. A limitation of the algorithm is the use of device-specific MAC addresses for dynamic key derivation. This method increases adaptability and uniqueness but can be problematic. The issues arise in environments where MAC addresses are easily spoofed or changed. This change can undermine the security model. Another challenge that the algorithm faces is the increasing heterogeneity of IoT ecosystems. Ensuring seamless interoperability and backward compatibility with legacy systems is complex. This complexity increases as either device standards or network protocols evolve. Although scalability is already a consideration in the current design, new bottlenecks for IoT networks scaling to billions of devices and data volumes increase concerns. Memory access patterns and computational overhead for ultraconstrained devices at extreme scales must be empirically authenticated as a further refinement of the algorithm performance.

Ultimately, side-channel attacks remain a concern, for example, timing, electromagnetic, or power consumption-based attacks. NEHA offers state-of-the-art block reordering and dynamic keying, but further attack countermeasures are necessary to create resilient defenses against ever more intricate side channel attacks. NEHA on its own provides an extremely well-rounded foundation for secure IoT communications, but work should target quantum resistance, improved interoperability, large-scale performance verification so that a security margin for evolving enabling technologies remains, and enhanced side-channel protections to enable/guarantee useful security and adaptability in newly emerging network landscapes.

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Abhiram Sharma (Member, IEEE) is currently pursuing the B.Tech degree in computer science and engineering from VIT, Chennai, India, specializing in data science.

With a strong foundation in computer vision, artificial intelligence, deep learning, and cybersecurity, he has consistently demonstrated excellence in academics, research, and leadership. His patents for novel designs reflect a passion for ideating and implementing impactful solutions. Driven by a commitment to pushing boundaries in innovation, he aspires to bridge the gap between theoretical advancements in technology and real-world applications, aiming to contribute significantly to the fields of secure communication, AI, and IoT technologies. As an author of multiple innovative research initiatives, he has showcased expertise in designing efficient algorithms and utilizing advanced technologies such as transfer learning and convolutional neural networks. Beyond technical expertise, he has significantly contributed to the university's academic community by founding the Engagements and Collaborations Committee of Data Science (ECDS). This initiative promotes student collaboration, organizes impactful technical events, and fosters cross-disciplinary engagement, making ECDS a hub for innovation and empowerment under his leadership. In addition to research, he has a proven track record of leadership and innovation, being recognized on an international stage as the Leader of Team India with FLL 2019 in Sydney, NSW, Australia. His research interests focus on the innovative application of AI in healthcare, agriculture, and environmental challenges, with notable projects including explainable AI (XAI)-based systems for medical diagnostics and anomaly detection.

Mr. Sharma won the Best Project Award with FLL 2019.



Rudra Raj (Member, IEEE) is currently pursuing the B.Tech degree in computer science and engineering from Vellore Institute of Technology, Chennai, India, specializing in AI and Robotics.

He has always been enthusiastic about new and advanced cryptography skills.

Mr. Raj is a member of the Robotics Club of VIT Chennai.



P. B. Krishna (Member, IEEE) is currently pursuing the B.Tech degree in computer science and engineering from Vellore Institute of Technology, Chennai, India.

He maintains an academic record with a CGPA of 9.67. His academic and professional interests focus on deep learning, cybersecurity, and natural language processing, where he turns innovative ideas into solutions. He has co-authored multiple papers, including the development of a novel network-efficient hierarchical encryption algorithm for enhancing IoT encryption. His technical portfolio extends beyond research, with projects such as Lumi Dissect, a surgical tool considered to have streamlined treatment. He has also used machine learning models in language translation for underrepresented language pairs. In addition to his technical expertise, he is serving as the Head of Content for the ACM Student Chapter at VIT Chennai. He has organized outreach programs, engaged audiences through technical content, and promoted coding literacy among students.

Mr. Krishna earned the prestigious Best Paper Award at an international conference.



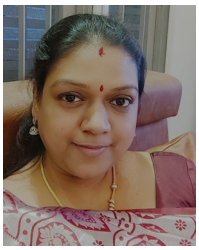
Ashwin Singh Slathia (Member, IEEE) is currently pursuing the B.Tech degree from Vellore Institute of Technology, Chennai, India, specializing in computer science and engineering.

He specializes in artificial intelligence, machine learning, and deep learning. He has contributed significantly to multiple papers of different domains, including CHASHNIt: Enhancing Skin Disease Classification leveraging GAN-Augmented Hybrid Model with Comparative XAI Interpretation Heatmap Using LIME and SHAP Algorithms and Remote Patient Monitoring. He has also published patents on fog prediction and real-time monitoring systems for aviation and urban applications, road collision avoidance, and omnidirectional drone-mounted nozzle extension for $360^\circ \times 180^\circ$ spraying applications. His research interests include machine learning and innovations, which continue to motivate him to pursue more such endeavors.



M. Afzal Khan (Member, IEEE) is currently pursuing the degree in computer science and engineering from VIT, Chennai, India, specializing in data science.

With strong technical expertise in full-stack web development, machine learning, and deep learning, with a keen interest in cyber security, he has built a diverse portfolio of projects that combine practical utility with innovative design. He has developed numerous websites like Econet Infinity, OMR Homes, and so on. He also has tailored AI-based tools for both educators and personal use and translated complex ideas into functional and user-friendly applications. He is also a member of the CodeChef Club, VIT Chennai, as a competitive programmer.



R. Suganya (Member, IEEE) received the bachelor's degree (Hons.) in engineering from Madurai Kamaraj University, Madurai, India, in 2003, specializing in Computer Science and Engineering, the master's degree (Hons.) from Anna University, Chennai, India, in 2005, and the Ph.D. degree from Thiagarajar College of Engineering, Madurai, in 2014, focusing on "classification and retrieval of ultrasound liver images using machine learning algorithms."

She boasts 18 years of expertise in computing, predominantly within academic institutions in South Tamil Nadu, India. Currently, she serves as an Associate Professor Senior with the School of Computer Science and Engineering, VIT, Chennai. She previously dedicated 16 years to Thiagarajar College of Engineering, Madurai. Her academic journey notably includes the following: she completed an industry consultancy project with Alfa TKG, Japan, in 2021, and received VIT SEED and travel grants from the India–Sri Lanka Foundation, aiming to share research experiences in Information Security and Blockchain Technology with scholars in Jaffna, Sri Lanka. As a prolific researcher, she has authored 11 book chapters published by reputable publishers. Her research portfolio includes over 75 articles covering diverse areas such as medical imaging, obstacle detection, sclera segmentation, AI-based paper evaluation, and smart translators for the visually impaired. These contributions have found homes in esteemed journals like the *Journal of Intelligent Fuzzy Systems*, *Journal of Applied Artificial Intelligence, Cluster Computing*, and others. Beyond research, she serves as a reviewer for various journals and has delivered over 50 technical lectures in national seminars, faculty development programs (FDPs), workshops, conferences, and special lectures on AI. She holds three patents, including one for "head gear for stress controller." In the academic year 2021–2022, she assumed the role of an Associate Dean for International Relations, fostering collaborations with prestigious institutions worldwide. Her research interests span medical imaging, artificial intelligence, natural language processing, computer vision, big data technologies, software engineering, the Internet of Things, and engineering education.

Dr. Suganya's outstanding contributions have earned her prestigious awards, including the ACM Women's chapter Emerging Chapter Award in 2024, the Best Familiar Faculty Award from 2016 to 2017, and the Young Women in Engineering Award in 2017. She has also been recognized as the best mentor in Women in Big Data (WiBD) in 2023.



Nabin Sharma (Senior Member, IEEE) received the Ph.D. degree from the School of ICT, Griffith University, Brisbane, QLD, Australia, in 2015.

He is currently an Associate Professor with the School of Computer Science, Founder and Co-Director of Intelligent Drone Lab, Faculty of Engineering and IT, University of Technology Sydney, NSW, Australia. His research area focuses on video and image processing, pattern recognition, and machine learning techniques for object detection and recognition. He has more than 18 years of experience in research and development and academia. He has substantial industry experience in software design and development while working on various projects at IBM India Private Ltd. He has published more than 60 articles in refereed books, conferences, and journals. His Google Scholar profile reports 2318 citations with an i10-index of 35 and an H-index of 25. He secured research grants for projects with funds exceeding AUD1.7M+, in collaboration with industry and academia.

Dr. Sharma received the runner-up award in "The Young IT Professional Award Competition 2006" organized by the Computer Society of India (CSI), East Zone, and was also nominated for the prestigious "Young Scientist Award 2006," of the Indian Science Congress Association. He received the "Spirit of GovHack 2015" award at GovHack 2015, Griffith University Gold Coast Campus. He was the winner of iAward NSW 2018 in the following three major categories for the SharkSpotter project: 1) Research and Development Project of the Year, 2) Artificial Intelligence or Machine Learning Innovation of the Year, and 3) Community Service Markets. The Australian Information Industry Association (AIIA) NSW iAwards is the nation's leading awards program for innovation in the digital economy. He was the winner of the National iAwards 2018 for the SharkSpotter project in the Artificial Intelligence or Machine Learning Innovation of the Year category. He received the Merit Award at the Asia Pacific ICT Alliance Awards (APICTA) 2018 for the SharkSpotter project in the Artificial Intelligence Technology of the Year category. He won the Australian Association for Unmanned Systems (AAUS) industry awards 2020, innovation champion, for the ultralow latency live streaming from UAVs and SpotterAI© suite, along with the Ripper Group Pty. He was the winner of the 2020 Queensland State iAward for the BotVision project, in the start-up of the year category, along with Codebots.